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Instrumental and integrative motivations predict intention to use generative AI chatbots for English learning among undergraduates with non English majors in Chinese private universities

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Abstract

Purpose This study integrates Gardner's Socio-Educational Model of Second Language Motivation (SEMSLM) with the Technology Acceptance Model (TAM) to examine the influence pathways of instrumental motivation (INS) and integrative motivation (INT) on Chinese non-English major undergraduates' intentions to use generative AI (Gen-AI) chatbots.

Methodology A mixed random and convenience sampling approach was employed to survey students from four private universities in China, yielding 729 valid responses. The study measured students' instrumental and integrative motivation, as well as TAM constructs including perceived usefulness (PU), perceived ease of use (PE), and behavioral intention to use (IU). Structural equation modeling (SEM) was applied to analyze both direct and indirect effects of motivation on IU, investigating the mediating mechanisms of TAM variables.

Findings Participants demonstrated a clear preference for domestically developed Gen-AI chatbots and reported relatively high instrumental motivation and perceived ease of use. Instrumental motivation significantly influenced IU through three mediation paths, with a stronger effect via PE than PU. In contrast, integrative motivation influenced IU solely through PU. Overall, the total effect of instrumental motivation was substantially higher than that of integrative motivation.

Value The study highlights differential pathways through which distinct motivational types shape Gen-AI chatbot adoption, providing insights for supporting low-motivation learner groups and advancing understanding of AI-assisted language learning.

Keywords Generative AI (Gen-AI) chatbot, Non-English-major undergraduates, Private universities in China, Socio-Educational Model of Second Language Motivation (SEMSLM), Structural equation modeling (SEM), Technology Acceptance Model (TAM)

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1 Introduction

With the widespread application of generative artificial intelligence (Gen-AI) chatbots—exemplified by ChatGPT—students have been shown to leverage these tools to improve academic outcomes [1], and similar benefits have been documented in English-language learning contexts [2, 3]. Yet concerns over academic dishonesty have prompted some institutions to prohibit their use [4], and existing studies have largely focused either on usage patterns and feature optimization [5] or on misuse and regulatory responses [6]. Because Gen-AI is widely recognized as a major trend in educational technology [7], a constructive, long-term integration of these tools within learning environments is preferable to purely restrictive approaches [8–10]. In China, where foreign-language teaching increasingly emphasizes positive-psychology factors [11–13], understanding the motivational determinants of Gen-AI use—especially in extracurricular digital English-learning contexts—remains critical, yet under-researched [14].

Research on Chinese undergraduates' use of Gen-AI chatbots for language learning has predominantly centered on ChatGPT [15–19], overlooking domestically available platforms that are widely adopted in mainland China due to bans on certain foreign services [20]. This mismatch between the research focus and students' actual tool preferences limits our understanding of how and why learners choose specific Gen-AI environments, making it necessary to clarify the distribution of platform preferences in China's current digital ecosystem.

Non-English major undergraduates at private universities in China often face structural challenges, such as weak English proficiency, exam-oriented curricula, low intrinsic interest, and negative emotional experiences such as anxiety and avoidance. These factors significantly undermine their motivation and self-efficacy for language learning [21–23]. Despite evidence that Gen-AI tools open new possibilities for language learning, and that self-efficacy influences both students' intention to use such tools [24] and their learning motivation [25], Yang and Chen [26] found that students with low motivation may lack enthusiasm or even exhibit resistance toward English-related learning, including the use of Gen-AI tools for language learning. This suggests that students with low motivation are unlikely to develop strong intentions to use Gen-AI tools for learning English via the Technology Acceptance Model (TAM) pathways. Moreover, few studies have examined how motivational characteristics influence the adoption of Gen-AI technology among a broader range of learners, and little is known about the motivational profiles of students in private universities.

Extant literature has linked Self-Determination Theory [27], the L2 Motivational Self System [28], and other motivational constructs [8, 29, 30] with students' intentions to use AI tools via the Technology Acceptance Model's core variables—perceived usefulness (PU) and perceived ease of use (PE) [6, 16, 31–35]. However, Socio-Educational Model of Second Language Learning (SEMSLM) proposed by Gardner [36, 37], with its focus on instrumental and integrative motivation, has not been fully integrated into TAM in this context. Although Wang [38] demonstrated that both instrumental and integrative motivations can directly influence the use of mobile apps and Gen-AI tools for language learning, the structural interplay between these motivational drives and TAM pathways remains unexplored. Theoretically, we further examine the distinct mechanisms of the two types of motivation through explicit causal pathways. Should the empirical results support these paths and their interaction effects, the findings would encourage the TAM

framework to treat motivational sources as structured antecedent variables in explaining technology adoption. Simultaneously, this would extend the SEMSLM model to technology-mediated language learning contexts, elucidating how motivation influences technology adoption via perceptual mechanisms.

To address these gaps, the present study investigates non-English-major undergraduates at private Chinese universities within an Informal Digital Learning of English (IDLE) [39] context, thereby minimizing potential influences from classroom instruction and environment and authentically examining spontaneous psychological engagement with Gen-AI chatbots [40]. The study integrates instrumental and integrative motivation from SEMSLM with PU, PE, and intention to use (IU) from TAM to construct and validate a comprehensive model of Gen-AI adoption behavior.

This study makes three key contributions. First, it focuses on non-English major undergraduates in Chinese private universities and incorporates their actual preferences for Gen-AI platforms, thereby providing highly contextualized empirical evidence grounded in the Chinese setting. Second, the study addresses a research gap by systematically integrating Gardner's SEMSLM [36, 37] with the TAM [41]; although previous studies exist, few have jointly examined the roles of instrumental and integrative motivation within the PU/PE pathways. Third, the study not only extends the explanatory power of motivational variables in technology adoption models but also offers a valuable complement and challenge to prior research that primarily relies on theories such as SDT [27]. Specifically, based on the foregoing, this study addresses the following research questions:

1. What are students' brand preferences when using Gen-AI chatbots for English learning?
2. What are the overall levels of instrumental motivation, integrative motivation, PU, PE, and IU among this cohort?
3. What is the structural relationship among motivational factors, PE, PU, and IU?

2 Literature review

2.1 Technology Acceptance Model (TAM)

Davis [41] defined perceived ease of use (PE) as the effort one expects to expend when using a new technology and perceived usefulness (PU) as the degree to which one believes the technology will enhance task performance; both predict intention to use (IU), with PE also influencing PU. Subsequent extensions of TAM have added antecedents while retaining PE and PU as mediators of IU [42–44]. Its robustness is affirmed across domains—including education and emerging Gen-AI research [16, 31, 45, 46]—but Kanont et al. [47] found among Thai undergraduates that PE did not affect IU and even negatively correlated with PU. Moreover, Wu et al. [48] found that PU did not significantly influence Chinese university students' intention to use Gen-AI for English learning, challenging the traditional TAM pathways. Overall, existing research generally supports the applicability of the TAM framework in educational and AI contexts; however, it also indicates that the relationships among its core constructs may vary depending on the technology and context, leaving room for further investigation. Accordingly, under IDLE conditions we hypothesize:

H1: PE of Gen-AI will have a positive effect on PU.
H2: PE will have a positive effect on IU.

H3: PU will have a positive effect on IU.

2.2 Motivation and intention to use

Recognizing that TAM may oversimplify motivational antecedents [49] and that empirical evidence on language learners' educational-technology use is limited [50, 51], this study introduces motivation as an antecedent variable in the IDLE context to extend the TAM framework in a Gen-AI–assisted English-learning environment.

In the field of second language acquisition (SLA), Gardner [36, 37] proposed two core types of motivation within Socio-Educational Model of Second Language Motivation (SEMSLM): instrumental motivation and integrative motivation. Instrumental motivation refers to learners' utilitarian purposes for acquiring a language, such as improving test performance, advancing in their careers, or fulfilling specific tasks. This type of motivation closely aligns with the concept of external motivation, which emphasizes the role of external rewards or punishments in shaping behavior [27]. Previous research has confirmed that this motivational orientation can influence learners' behavioral intentions toward the use of generative artificial intelligence (Gen-AI). For instance, Liu and Darvin, et al. [52] examined the role of motivation in Chinese university students' engagement in AI-mediated informal digital learning of English (AI-IDLE). They found that the "ideal L2 self," a construct that includes instrumental motivation—such as "Whenever I think of my future career, I imagine myself using English"—directly influenced learners' AI-IDLE intentions.

Integrative motivation also has been shown to predict the sustained use of language learning applications [53]. It bears similarity to intrinsic motivation, which refers to engagement in an activity driven by inherent interest, enjoyment, or satisfaction, rather than external outcomes [27]. In fact, integrative motivation is more domain-specific and socially oriented and arises from the learner's desire to identify with and become part of the target language community [36, 37]. This motivation is maintained through a sense of cultural affiliation and interaction with members of the target language group. Originally, the concept of integrative motivation was developed within multilingual contexts [36, 37]—such as immigrant-receiving countries like the United States—where authentic interaction with a target-language community is possible. However, cross-cultural research by Taguchi et al. [54] across Japan, China, and Iran has shown that integrative motivation retains significant explanatory power even in the absence of real-life English-speaking communities. This suggests that the influence of integrative motivation is culturally transferable and not contingent on the actual presence of the target language community. Furthermore, many Gen-AI tools (e.g., AI conversation partners, cultural simulation modules) now simulate immersive target-language cultural environments [55]. These technological affordances allow learners to experience cultural presence virtually, thereby activating and sustaining their integrative motivational orientation.

Recent studies in AI-assisted language learning (AILL) further support the role of integrative motivation. For example, An et al. [33] investigated middle and high school students' intentions to use AILL technologies and identified a positive relationship between learners' cultural interest with AI—as reflected in items like "I like to learn and enjoy English movies and TV series through AI software"—and their willingness to engage with AILL tools. Although Wang [38] similarly examined the influence of different types of motivation on English as a Foreign Language (EFL) learners' intentions to use large

language models (LLMs), and found that both instrumental and integrative motivation significantly predicted IU, it is noteworthy that Sun and Gao [56], in their study on the impact of intrinsic motivation on students' intention to use mobile devices in mobile-assisted language learning (MALL), reported that intrinsic motivation did not directly influence IU. Furthermore, Yang et al. [26] noted that students with low motivation in learning English as a foreign language often lack enthusiasm for instructional and learning activities, which also extends to resistance toward AILL. Across the existing literature, both instrumental and integrative motivation have been shown to relate to learners' technology adoption; however, the direction and strength of these effects may vary across different contexts and populations, such as students with low motivational levels. This suggests that the pathway through which motivation affects intention to use technology, warrants further investigation. Based on the above findings, the current study proposes the following hypotheses in the context of IDLE:

H4: Instrumental motivation (INS) will have a direct effect on students' IU of Gen-AI chatbots for English learning.

H5: Integrative motivation (INT) will have a direct effect on IU.

2.3 PE and PU as mediators

Even prior to the advent of Gen-AI, the chain mediation effect of Motivation → PE→ perceived usefulness (PU) → intention to use (IU) had already been confirmed in non-educational contexts by Caniëls et al. [57]. This mediational pathway has also been supported in the field of technology-assisted language learning, as demonstrated by Hsu [58]. These studies consistently show that learning motivation predicts both PE and PU, which in turn mediate the relationship between motivation and learners' behavioral intention to use a given technology. For example, in a study exploring factors influencing Chinese secondary school students' use of an AI-powered tutoring system for English learning, Ni and Cheung [59] found that one motivational variable indirectly influenced IU via both the PU pathway and the PE → PU pathway, whereas another motivational factor affected IU only through the PU pathway.

In the Gen-AI era, this basic mediational structure appears to remain stable. For instance, Qu and Wu [17] found that motivational variables were significantly associated with IU, primarily through the mediating effect of PU. Similarly, Li and Zhang [16] reported that motivation indirectly influenced both PU and PE, with PU exerting a direct effect on IU, while PE influenced IU mainly through indirect pathways. However, it should be noted that these pathways may not be stable in English-as-a-second-language educational contexts. For example, Hsu [58] found a negative effect of motivation on perceived ease of use (PE), while Fan [60] suggested that motivation can directly predict intention to use (IU) rather than acting through perceived usefulness (PU). Additionally, when using Gen-AI tools, students may exhibit hesitation or uncertainty due to unfamiliarity or concerns about overreliance on technology [61]. Overall, the literature indicates that motivation generally influences IU indirectly through PE and PU, but the stability of these pathways varies, and findings are not entirely consistent across studies. Accordingly, under IDLE conditions this study posits (the hypothesized model is depicted in Fig. 1):

H6: Instrumental motivation (INS) will have a direct effect on PE.

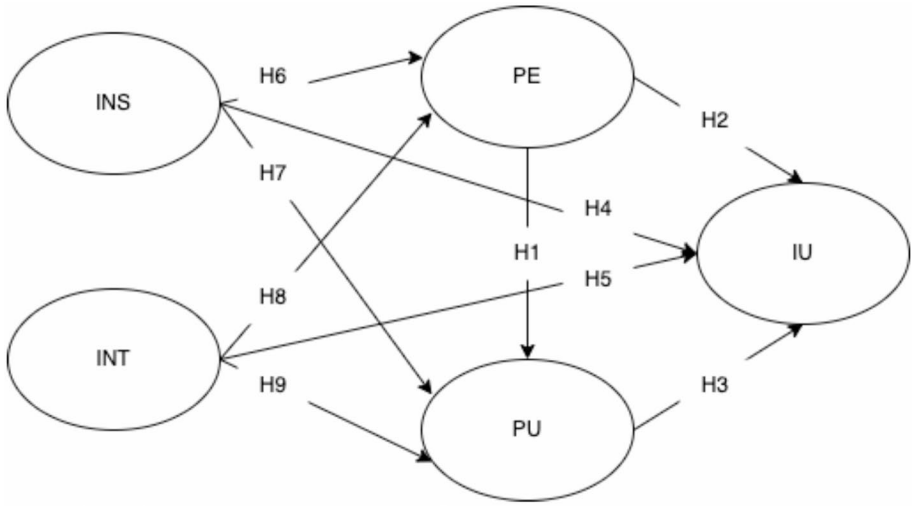

Fig. 1 Hypothesis model. *INS* instrumental motivation, *INT* integrative motivation, *PE* perceived ease of use, *PU* perceived usefulness, *IU* intention to use

H7: INS will have a direct effect on PU.

H8: Integrative motivation (INT) will have a direct effect on PE.

H9: INT will have a direct effect on PU.

3 Methodology

3.1 Participants

This study was conducted in Shandong Province, China, known for its strong private higher education sector and emphasis on internationalization and digitalization [62]. One private university was randomly selected from each of the province's four administrative regions, and 1000 online questionnaires were distributed through stratified convenience sampling proportional to student enrollment. Researchers and the colleagues visited the campuses at different times to invite students to complete the questionnaire via a QR code link to *Wenjuanxing*. The data collection period lasted from January 2, 2025, to April 30, 2025. Participants were informed about the study's purpose, confidentiality, and their right to withdraw without penalty, with written informed consent obtained. All research procedures were carried out in accordance with the Declaration of Helsinki.

To ensure data quality, the inclusion criteria were explicitly defined as follows: first- and second-year non-English-major undergraduates from the four universities (as these cohorts are in the compulsory College English course stage), aged 18 or above (to ensure full informed consent capacity in line with ethical requirements).

The exclusion criteria and rationale were as follows: English-major students (as their motivational profiles differ significantly from those of non-majors, which could confound the core variables); third- or fourth-year undergraduates (as they have completed the College English requirement, making it difficult to guarantee an ongoing English learning status); students under 18 years old (as parental consent would be required, complicating implementation, and more importantly, their cognitive maturity may differ from adults); and responses completed within 60 s (since, when the number of items exceeds 20, both the rate of "social desirability bias" and "random responding" exceeds

60% in such cases, necessitating direct exclusion; see [63]. After applying these screening procedures, a total of 729 valid questionnaires were retained for analysis.

3.2 Research instruments

The questionnaire comprised two parts. The first gathered demographics (gender, age, year, major) and Gen-AI brand preferences. The second—detailed in Appendix—included two validated scales (all $\alpha \geq 0.70$) [64].

The motivation scale, adapted from [65], contained ten 5-point Likert items: five for instrumental motivation (INS1–INS5) and five for integrative motivation (INT1–INT5). The research retained only Gardner's [36, 37] core SEMSLM dimensions—excluding classroom-specific subscales—to suit the IDLE context; despite differences from Nikitina et al.'s [65] Malaysian sample, maintaining all ten items ensured comprehensive coverage. Reliability and validity analyses confirmed its fit for non-English-majors in Chinese private universities. As the original scale targeted French language learning, all references to French were replaced with English. For example, INT4 "I decided to learn this language in order to better understand the French people's way of life" was adapted to "I decided to learn this language in order to better understand the English people's way of life," and INS3 "Proficiency in French can bring me some part-time job opportunities" was retained with "Proficiency in English can bring me some part-time job opportunities". The use and adaptation of this scale were authorized by the scale developer.

The TAM scale, based on Davis [41], consisted of fifteen 7-point Likert items: PU1–PU6 and PE1–PE6 from Davis's original instrument, plus three IU items (IU1–IU3) from Liu and Ma [40]. These IU items were selected because their original context closely aligns with the present study: Liu and Ma designed them specifically to measure university students' intention to use ChatGPT for English learning. To adapt the items for the broader Gen-AI context in IDLE, for instance, IU2 "I expect to use ChatGPT frequently in the future for learning English" was modified to "I expect to use Gen-AI chatbot frequently in the future for learning English."

Additionally, Davis's original PU items were formulated for general technology evaluation and did not explicitly reference English learning, which could lead to misunderstanding if used directly (e.g., students might conflate "technology usefulness" with "English learning usefulness"). Therefore, the PU items were adapted to specify English learning tasks; for example, PU1 "Using the Technology would enable me to accomplish tasks more quickly" was revised as "Using Gen-AI chatbot would enable me to accomplish English learning tasks more quickly." The use and adaptation of this scale were authorized by the original scale developer.

3.3 Data consistency and reliability

To ensure data consistency and reliability, the following procedures were implemented. First, content validation and questionnaire optimization were conducted. All adapted items were evaluated for content validity by five faculty members from Chinese universities (including two professors of applied linguistics and one associate professor of educational psychology), yielding CVI (Content Validity Index) values ≥ 0.88, indicating high content validity [66]. Moreover, to accommodate English as a foreign language learners, the questionnaire was translated into English and back-translated by two professional translators with English linguistics backgrounds; the back-translation consistency

coefficient reached 0.93, ensuring semantic equivalence before importing the survey into *Wenjuanxing*.

Second, informed consent and standardized instructions were implemented. Participants were required to read a written informed consent form and confirm their agreement before accessing the questionnaire. Each scale section included a uniform prompt, such as "This section assesses your perceptions of Gen-AI-assisted English learning," to avoid bias from in-person explanations.

Third, technical controls within *Wenjuanxing* were employed. Students accessed the survey by scanning the QR code with their personal WeChat accounts, each of which is unique, allowing the system to identify respondents and prevent multiple submissions by the same device. Responses with completion times under 60 s were automatically excluded to further ensure data quality.

Finally, a pilot study was conducted. Prior to formal data collection, 50 lower-year non-English major students were sampled from one university to complete the pilot survey. Pilot reliability analyses yielded Cronbach's $\alpha = 0.78$ for the motivation scale and $\alpha = 0.82$ for the TAM scale, meeting psychometric standards [64].

3.4 Data analysis

The collected data were analyzed using SPSS 29 and SmartPLS 4, applying the Partial Least Squares Structural Equation Modeling (PLS-SEM) approach. PLS-SEM is particularly effective for estimating complex causal relationships under conditions of small sample size or non-normal data distribution [67]. Following the two-step analytical strategy proposed by Anderson and Gerbing [68], the study first assessed the reliability and validity of the measurement model, and subsequently examined the significance and explanatory power of the structural paths in the model. This procedure ensured the robustness of the measurement instruments and the validity of the proposed research hypotheses.

4 Results

4.1 Demographic information and participants' brand preference

The demographic characteristics of the sample are presented in Table 1. The sample showed a generally balanced distribution in terms of gender, academic year, and academic field. In terms of gender, 59.3% were female and 40.7% were male. Regarding academic year, 68.2% were freshmen and 31.8% were sophomores. Most participants were aged between 18 and 19 (78.5%). As for academic field, the proportion of students in science and engineering (55.4%) and liberal arts (44.6%) was relatively close. Figure 2 shows the distribution of participants' preference for Gen-AI chatbot brands. Of the top ten

Table 1 Demographic characteristics of the sample

	Category	Number	Percentage (%)
Gender	Male	297	40.70
	Female	432	59.30
Grade	Freshman	497	68.20
	Sophomore	232	31.80
Age	18	225	30.90
	19	347	47.60
	20 and above	157	21.50
Academic field	Science and Engineering	404	55.40
	Liberal Arts	325	44.60

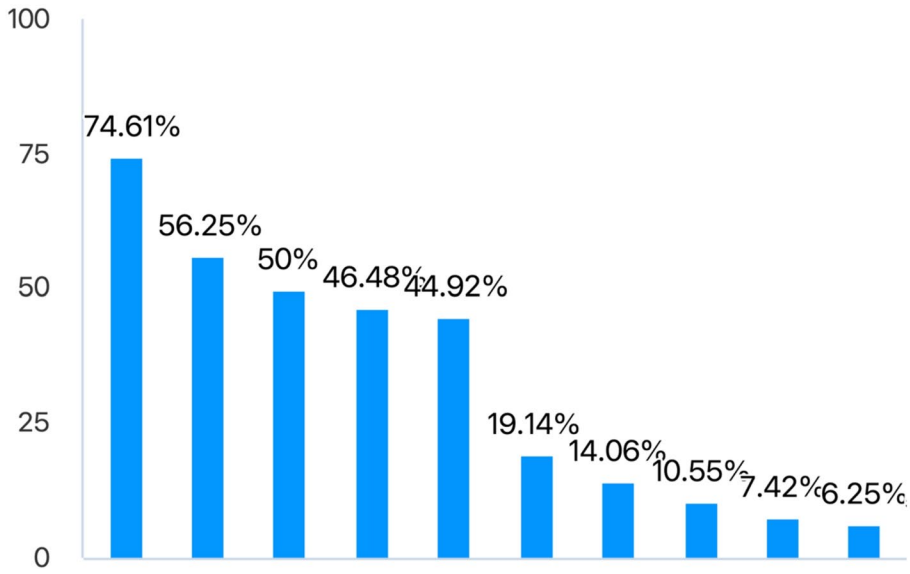

Fig. 2 Participants' preference of gen-AI chatbot brands. As this was a multiple-response question, the percentage for each option was calculated by dividing the number of times the option was selected by the total number of valid responses. Students' top ten preferred Gen-AI brands, in descending order, were: Doubao (74.61%), iFlytek Spark (56.25%), Kimi Chat (50%), Tongyi Qianwen (46.48%), Tiangong AI (44.92%), HunYuan Assistant (19.14%), Xiaomi MiLM (XiaoAi) (14.06%), ERNIE Bot (10.55%), ChatGPT (7.42%), Youdao Chat (6.25%)

brands most preferred by students, nine are domestic, and only ChatGPT, ranked ninth, is a foreign brand.

4.2 Descriptive statistics of constructs

As shown in Table 2, the item means for INS range from 4.050 to 4.290, indicating that participants generally endorse the utilitarian value of English learning. In contrast, INT means range from 2.830 to 3.930. The standard deviations for INS (0.847–0.995) and INT (1.027–1.123) reveal substantial inter-individual variability––some students hold very strong attitudes, whereas others are comparatively less inclined. Regarding the use of a Gen-AI chatbot for English learning, PE item means lie between 5.050 and 5.310, PU (perceived usefulness) between 5.260 and 5.430, and IU between 5.140 and 5.320, all reflecting high overall endorsement. However, whereas PU standard deviations (1.203–1.261) suggest relatively consistent views on usefulness, greater dispersion emerges for PE (SD = 1.329–1.467) and IU (SD = 1.350–1.385). All items across all constructs exhibit absolute values of skewness and kurtosis below 1, indicating no severe departure from normality; the data approximate a symmetric, normal distribution [69].

4.3 Structural equation modeling analysis

4.3.1 Measurement model

The measurement model assesses reliability and validity of the survey instrument [70]. First, following Hair Jr et al. [64], each latent variable's Cronbach's α (α) and Composite Reliability (CR) should exceed 0.70 to demonstrate adequate internal consistency. As summarized in Table 3, all constructs in this study had α and CR values ranging from 0.74 to 0.91, meeting the commonly accepted thresholds and demonstrating excellent internal consistency [64].

Table 2 Descriptive statistics

Construct	Item	Mean	SD	Skewness	Kurtosis
INS	INS1	4.110	0.995	−0.985	0.539
	INS2	4.110	0.867	−0.713	0.082
	INS3	4.200	0.847	−0.989	0.943
	INS4	4.290	0.851	−0.971	0.330
	INS5	4.050	0.897	−0.686	0.000
INT	INT1	2.830	1.120	0.077	−0.597
	INT2	3.620	1.053	−0.666	0.187
	INT3	3.720	1.110	−0.690	−0.098
	INT4	3.880	1.027	−0.949	0.659
	INT5	3.930	1.123	−0.996	0.404
PE	PE1	5.070	1.467	−0.456	−0.177
	PE2	5.050	1.433	−0.440	−0.094
	PE3	5.210	1.349	−0.424	−0.203
	PE4	5.140	1.395	−0.411	−0.370
	PE5	5.110	1.339	−0.353	−0.178
	PE6	5.310	1.329	−0.395	−0.385
PU	PU1	5.310	1.239	−0.246	−0.530
	PU2	5.260	1.203	−0.239	−0.328
	PU3	5.300	1.257	−0.467	−0.159
	PU4	5.320	1.214	−0.383	−0.098
	PU5	5.310	1.243	−0.298	−0.469
	PU6	5.430	1.261	−0.440	−0.452
IU	IU1	5.320	1.350	−0.552	−0.140
	IU2	5.250	1.366	−0.418	−0.295
	IU3	5.140	1.385	−0.286	−0.613

Convergent validity was evaluated via item loadings and average variance extracted (AVE). Except for four items—INS1 (0.466), INS4 (0.595), INT1 (0.618), and INT5 (0.686)—all outer loadings exceed the 0.708 benchmark. Hair Jr et al. [71] note that the presence of only one or two loadings below 0.708 is acceptable. Therefore, these four items were retained. Moreover, as shown in Table 3, all constructs had AVE values above 0.50, indicating that the latent variables explained more than half of the variance of their observed indicators, demonstrating good convergent validity [64].

Then, discriminant validity was examined using the Fornell–Larcker criterion and the heterotrait–monotrait ratio (HTMT). In Table 4, the square roots of each construct's AVE exceed its correlations with other constructs, meeting Fornell–Larcker's criterion [72].

In Table 5, all HTMT values remain below 0.90, further affirming discriminant validity [73]. These results indicate no high construct overlap and sufficient discriminant validity.

In summary, by comparing the observed metrics with established thresholds, the scales used in this study demonstrated satisfactory reliability, convergent validity, and discriminant validity. Specifically, all α and CR values exceeded 0.70, most outer loadings were above 0.708, AVE values were above 0.50, the square roots of AVE exceeded inter-construct correlations, and HTMT values were below 0.90, supporting the psychometric soundness of the instruments.

Table 3 Reliability and convergent validity

Construct	Reliability		Convergent validity		
	α	CR	Item	Loading	AVE
INS	0.745	0.789	INS1	0.466	0.505
			INS2	0.820	
			INS3	0.793	
			INS4	0.595	
			INS5	0.809	
INT	0.820	0.847	INT1	0.618	0.586
			INT2	0.839	
			INT3	0.826	
			INT4	0.830	
			INT5	0.686	
PE	0.914	0.915	PE1	0.821	0.701
			PE2	0.875	
			PE3	0.860	
			PE4	0.760	
			PE5	0.864	
			PE6	0.839	
PU	0.903	0.909	PU1	0.835	0.674
			PU2	0.820	
			PU3	0.887	
			PU4	0.857	
			PU5	0.779	
			PU6	0.739	
IU	0.812	0.813	IU1	0.847	0.727
			IU2	0.857	
			IU3	0.853	

Table 4 Discriminant validity by using the square root of AVE

	INS	INT	PE	PU	IU
INS	0.711				
INT	0.453	0.765			
PE	0.384	0.209	0.837		
PU	0.441	0.304	0.582	0.821	
IU	0.379	0.216	0.637	0.710	0.853

Table 5 Discriminant validity by using HTMT ratio

	INS	INT	PE	PU	IU
INS					
INT	0.600				
PE	0.447	0.229			
PU	0.533	0.349	0.638		
IU	0.481	0.254	0.741	0.819	

4.3.2 Structural model

To assess overall model fit, this study reports the standardized root mean square residual (SRMR) and the normed fit index (NFI), shown in Table 6. The SRMR of 0.060 falls below commonly recommended thresholds of 0.08 or 0.10.

SRMR measures the average deviation between predicted and observed covariances, with smaller values indicating closer alignment; thus, SRMR below 0.08 or 0.10 suggests minimal residuals and good model fit [71]. The NFI was 0.837, approaching 1. NFI

Table 6 Model fit

	Saturated model	Estimated model
SRMR	0.060	0.060
d_ULS	1.186	1.186
d_G	0.398	0.398
Chi-square	1734.859	1734.859
NFI	0.837	0.837

Table 7 Hypothesis testing direct effects

Hypothesis	Path	B	STDEV	t	*p*	Result
H1	PE→PU	0.479	0.037	12.864	< 0.001	Accept
H2	PE→IU	0.332	0.042	7.986	< 0.001	Accept
H3	PU→IU	0.507	0.039	12.995	< 0.001	Accept
H4	INS→IU	0.039	0.031	1.267	0.205	Reject
H5	INT→IU	− 0.025	0.027	0.916	0.360	Reject
H6	INS→PE	0.364	0.042	8.769	< 0.001	Accept
H7	INS→PU	0.207	0.042	4.953	< 0.001	Accept
H8	INT→PE	0.044	0.040	1.100	0.271	Reject
H9	INT→PU	0.110	0.036	3.078	< 0.005	Accept

evaluates the improvement of the fitted model relative to a null model (assuming no relationships among variables); values closer to 1 indicate greater improvement, suggesting acceptable fit in this study [74].

Three hypotheses (H4, H5, H8) were not supported ($p > 0.05$), whereas the remaining six hypotheses achieved statistical significance (see Table 7). Regarding the path coefficients (β values), PE significantly and positively influences both PU ($\beta = 0.479$, $p < 0.001$) and IU ($\beta = 0.332$, $p < 0.001$), indicating that perceived ease of use not only enhances perceived usefulness but also directly drives behavioral intention. PU exerts the strongest effect on IU ($\beta = 0.507$, $p < 0.001$), highlighting its central role in shaping usage behavior.

For the motivational constructs, INS has a significant positive effect on PU ($\beta = 0.207$, $p < 0.001$) but not on IU directly ($\beta = 0.039$, $p = 0.205$), suggesting that extrinsic motivation primarily influences IU indirectly through PU and PE. INT exhibits a weak yet significant positive effect on PU ($\beta = 0.110$, $p < 0.005$) but shows no significant direct effects on IU or PE, implying that the impact of intrinsic motivation also relies on mediating variables.

Overall, these findings indicate that PU and PE serve as crucial mediators linking motivational constructs to IU, with PU playing a particularly pivotal role; the effects of motivation on behavioral intention are largely indirect rather than direct.

Figure 3 also illustrates the structural model's standardized path coefficients and their associated p-values. The coefficients of determination (R^2) shown in Fig. 3 indicate the proportion of variance in each endogenous construct explained by its exogenous predictors, reflecting the explanatory power of the model [75]. Specifically, PE ($R^2 = 0.147$) exhibits weak explanatory power, PU ($R^2 = 0.401$) moderate, and IU ($R^2 = 0.578$) strong, consistent with the guidelines of Hair et al. [75] whereby $R^2 \geq 0.50$ is considered strong, 0.30–0.50 moderate, and 0.20–0.30 weak.

Overall, SRMR, NFI, and R^2 all met recommended criteria, and the combination of path coefficients and hypothesis testing indicates that the structural model exhibits satisfactory fit, with meaningful explanatory power for endogenous constructs.

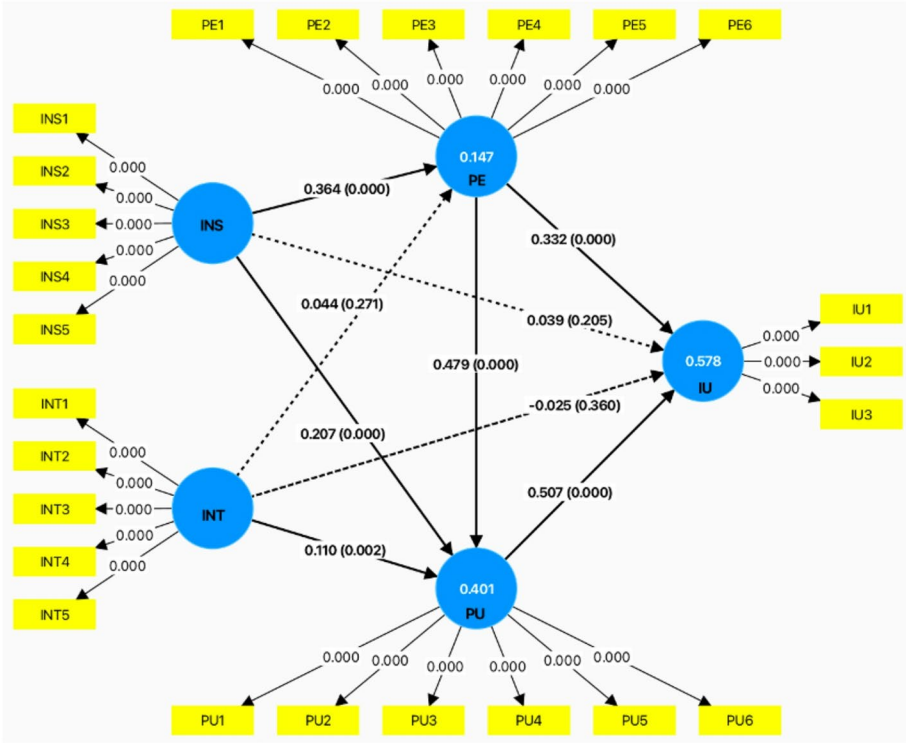

Fig. 3 Results of the structural model

Table 8 Mediation testing indirect effects

Indirect path	Direct effect	Indirect effect	Total effect	*p*	Result
INS→PE→IU	0.039	0.121	0.160	< 0.001	Accept
INS→PE→PU→IU	0.039	0.089	0.249	< 0.001	Accept
INS→PU→IU	0.039	0.105	0.144	< 0.001	Accept
INT→PE→IU	− 0.025	0.015	− 0.010	0.284	Reject
INT→PE→PU→IU	− 0.025	0.011	0.001	0.282	Reject
INT→PU→IU	− 0.025	0.056	0.031	< 0.005	Accept

4.3.3 Mediation analysis

To further investigate the underlying mechanisms linking INS and INT with IU, a mediation analysis was conducted to assess PE and PU as potential mediators. Following the procedure outlined by Nitzl et al. [76], bootstrapping within a PLS-SEM framework was employed to evaluate the significance and direction of both direct and indirect effects. As shown in Table 8, the mediation analysis confirmed that both PE and PU serve as mediators in the relationships between the two motivational variables and IU. Specifically, INS was found to predict IU via PE (total effect [TE] = 0.160) and via PU (TE = 0.144) independently, and also through the sequential mediation pathway INS → PE → PU → IU (TE = 0.249). In contrast, INT predicted IU only through PU (TE = 0.031). These findings provide a detailed basis for subsequent discussion.

These findings provide a detailed basis for subsequent discussion. Comparisons of direct and indirect effects indicate that INS has a more complex mechanism, involving both direct and sequential mediation through PE and PU, highlighting that extrinsic motivation influences IU by affecting both perceived ease of use and perceived usefulness. In contrast, INT operates primarily through PU, suggesting that intrinsic

motivation is translated into behavioral intention mainly via perceived usefulness. Overall, the analysis further validates the bridging role of PU and PE between motivation and IU, and illustrates the differential pathways of extrinsic versus intrinsic motivation.

5 Discussion

To address the first research question, Fig. 2 illustrates that among Chinese undergraduates in this study, brand preferences for Gen-AI chatbots in English learning exhibit a pronounced localization trend. Although the extant literature has largely concentrated on ChatGPT [15–19], reflecting a Western-centric sample and tool bias, in the present study, Doubao, a product of ByteDance, ranked first among surveyed students, whereas ChatGPT placed ninth. Nine of the top ten brands originate from domestic internet companies in China. This finding suggests that the future studies should pay greater attention to the localized dimensions, incorporating the "level of domestic technological development" into study design to avoid overgeneralizing brand preferences from a single cultural context, thereby enhancing both the external validity and practical relevance of the conclusions [77].

Regarding the second research question, descriptive statistics for each construct (in Table 2) reveal that INS scored relatively high (M = 4.15, SD = 0.89), suggesting that outcomes (e.g., examinations, further study, or career advancement) strongly drive students. In contrast, INT scored lower (M = 3.60, SD = 1.09) with substantial inter-individual variability, indicating that a relatively limited interest among students in the language itself or in cultural integration, accompanied by substantial individual differences in this dimension—an observation consistent with prior findings [21–23]. However, this result should not be simply attributed to students' "instrumental orientation," but rather interpreted as a motivation imbalance under contextual constraints. According to SDT [27], the activation of INT requires the fulfillment of three basic psychological needs: autonomy, competence, and relatedness. Students at private universities are generally constrained by exam pressure, insufficient self-efficacy, and limited cultural exposure, which hinder the expression of INT. In contrast, externally regulated INS is strongly activated in this context. Furthermore, the lack of cultural immersion features in current Gen-AI tools further suppresses the development of INT. Notably, the high variance observed in INT does not indicate weak motivation, but rather a "latent activatability." Lave and Wenger [78] and Berlyne [79] suggest that some students' INT remains dormant merely due to the absence of cultural participation opportunities; once appropriate learning contexts or tool functionalities are provided, these students can transition from "peripheral participation" to "deep engagement."

In addition, students' perceptions of Gen-AI-assisted English learning were also positive: perceived ease of use (PE) averaged 5.15 (SD = 1.39), perceived usefulness (PU) averaged 5.32 (SD = 1.24), and intention to use (IU) averaged 5.24 (SD = 1.37), which suggests strong acceptance of the technology's capacity to enhance the learning experience, albeit with individual differences in user experience. Considering the high levels of instrumental motivation and perceived ease of use reported in Table 2, utilitarian, exam-oriented learners typically initiate Gen-AI chatbots use based on operational convenience but remain skeptical about its long-term benefits for language proficiency [53, 59, 80]. At a fundamental level, this can be interpreted through the lens of Cognitive Load Theory (CLT) [81]: while high PE reduces barriers to adoption, it may also encourage

"surface-level engagement." PU, although acknowledged, primarily reflects the value for short-term tasks rather than long-term skill development. This suggests that without proper instructional guidance, Gen-AI tools risk being reduced to "instrumental answer-generation tools," rather than serving as vehicles for sustained language learning.

To answer the last research question, four main observations emerge. First, the hypothesized model and its evaluation results (Fig. 3; Table 7) indicate that, among the TAM variables, PE exerts a significant effect on IU ($\beta = 0.332$, $p < 0.001$), and PU exerts an even stronger effect on IU ($\beta = 0.507$, $p < 0.001$). Additionally, PE influences IU indirectly via PU (PE → PU, $\beta = 0.479$, $p < 0.001$). These findings align with [16, 18, 60], thus confirming TAM's core assumption that cognition precedes behavior, and that perceived ease of use is an integral component of perceived usefulness [41].

It is noteworthy that, within TAM, PU exerts a stronger influence on IU than PE, which reflects the contextual characteristics of non-English-major undergraduates at private Chinese universities—namely, high exam pressure coupled with low language self-efficacy [21–23]. Given their relatively weak English proficiency, these students perceive Gen-AI as a "crucial tool for improving scores" and prioritize whether it can enhance performance (PU) over ease of use (PE); even if the tool is somewhat complex to operate, they will still engage with it as long as it contributes to score improvement. This extends the application of TAM to educational contexts: in high-difficulty task scenarios such as language learning, performance-related usefulness (PU) drives IU more strongly than in general technology contexts, aligning with TAM2's perspective that "task importance moderates the PU–IU relationship" [82].

Second, findings indicate that in a model structure where PU and PE serve as strong predictors, the influence of motivation on IU may be fully mediated, thereby obscuring the statistical significance of its direct path. Specifically, neither INS nor INT directly predicted IU, leading to the rejection of Hypotheses 4 and 5. This result contrasts with [33, 38, 52, 53], but partially aligns with Sun and Gao [56]. A comparison of path models suggests that this discrepancy may stem from whether mediating variables—such as PU or PE—were included between motivational constructs and IU. In the present study, both PU and PE were incorporated as mediators between motivations and IU, and both demonstrated strong predictive effects on IU (PE: $\beta = 0.332$, $p < 0.001$; PU: $\beta = 0.507$, $p < 0.001$), which likely diverted the explanatory power of motivation, resulting in a full mediation effect.

The finding that INS and INT do not directly predict IU does not imply that motivation is irrelevant; rather, it reveals the "educational technology decision chain." As a learning tool, Gen-AI usage requires passing through the sequence of "motivation → perceived tool suitability (PU/PE) → intention formation." Even strong motivation cannot drive use if the tool is perceived as ineffective (low PU) or difficult to use (low PE). This aligns with SDT [27]: extrinsic motivation (e.g., INS) must be mediated by perceived tool fit to translate into behavior, whereas intrinsic motivation (e.g., INT) requires the tool to satisfy basic psychological needs [83]. The strong effects of PU and PE thus serve as a necessary bridge between motivation and IU, which also explains why studies that did not incorporate these mediators [33, 38, 52, 53] reported differing results—the discrepancies reflect differences in model specification rather than true contradictions in findings.

However, this complete mediation is not necessarily observed with other types of mediators. For instance, Liu and Darvin [52] introduced Enjoyment as a mediator between motivation and AI-IDLE. While the mediation effect of Enjoyment was statistically significant ($\beta = 0.31$, $p < 0.001$), its relatively limited effect size failed to fully mediate the relationship between motivation and AI-IDLE. Similarly, Wang et al. [53], in their investigation of factors influencing the intention to use language learning applications, found that Flow partially mediated the relationship between integrative motivation and IU. Yet, due to the relatively weak influence of Flow on IU ($\beta = 0.204$, $p < 0.001$), the mediating effect was insufficient to override the direct effect of motivational variables. Therefore, not all mediators can fully "block" the direct influence of motivation on behavior; the extent of full mediation depends on the strength and effectiveness of the mediating variables themselves.

Third, further path analyses revealed that INS influenced IU through three distinct mediating pathways: (1) INS → PE → IU, as supported by [16]; (2) INS → PU → IU, consistent with the findings of [59]; and (3) INS → PE → PU → IU, also corroborated by [58, 59].

These findings indicate a significant difference in the mechanisms of INS and INT. The effect pathway of INS exhibits multiple overlapping effects. For students with strong INS, their learning goals are highly utilitarian, reflecting a dual demand for both convenience and effectiveness: they value the time-saving and reduced cognitive load afforded by ease of use (PE), consistent with Cognitive Load Theory, while also emphasizing the assurance of achieving learning objectives (PU), in line with the concept of task importance in TAM2 [82]. This even manifests as a chained "PE→PU" effect, highlighting the efficiency-oriented nature of instrumental motivation [36, 37].

In contrast, INT influenced IU solely through the INT → PU → IU pathway, as Hypothesis 8 was rejected. Specifically, the standardized path coefficient from INT to PU was 0.110 ($p < 0.001$), while no significant effect was observed on PE—consistent with the findings of [56]. This suggests that, for the current sample, integrative motivation—rooted in interest in the target language and culture—primarily enhances attention to the instrumental utility (PU) of Gen-AI tools, rather than their operational ease (PE). For students with stronger intrinsic motivation, who have a genuine interest and passion for English and its associated culture, the driving force for learning does not depend on the tool's ease of use (PE), but rather on whether the tool can provide meaningful opportunities for cultural understanding and language deepening.

Nonetheless, the effect of INT on IU via PU remained modest (TE = 0.031, $p < 0.001$), indicating that the limited effect is not due to a lack of cultural interest, but rather that current Gen-AI tools are insufficient to meet these higher-order cultural and language learning needs. This finding aligns with the perspective from Situated Learning Theory [78] that "motivation cannot be activated in the absence of culturally rich learning contexts."

Interestingly, both instrumental and integrative motivation exerted their influence on IU indirectly via PU. This finding contrasts with Fan [60] and Wu et al. [48], who found that motivation directly predicted IU without mediation through PU. Fan attributed this to an exam-oriented educational context, where students did not perceive platforms such as WeChat or Bilibili as useful for improving English performance. By contrast, the

present study found that, regardless of motivational type, students generally acknowledged the practical utility of Gen-AI tools in English learning (M = 5.32).

The attenuation of integrative motivation may be associated with the global multipolar trend, which has weakened learners' drive to integrate into target-language communities [84], compounded by Chinese students' immersion in a Chinese-dominant linguistic environment that reduces the appeal of cultural integration—a phenomenon also linked to China's unique Internet ecosystem [85]. Nonetheless, from a long-term learning perspective, cultural engagement serves as a critical driver for deepening target language comprehension, cultivating higher-order interests, and sustaining ongoing learning [86]. According to Deci and Ryan [87], embedding authentic cultural content within Gen-AI can effectively activate intrinsic motivation. Empirical evidence indicates that, compared with traditional instrumentally driven processes, this approach has a more pronounced effect on enhancing integrative motivation [88]. Moreover, consistent with Deci and Ryan [87], when this motivation is activated, it can also improve students' emotional states, eliciting more positive emotions and thereby promoting greater focus and engagement in academic tasks.

Fourth, the path coefficients from INS to PE and PU were 0.364 and 0.207 respectively ($p < 0.001$), indicating that goal-oriented motivation places greater emphasis on lowering usage barriers and reducing cognitive load than on the perceived usefulness of the tool per se. This aligns with Gardner's [36, 37] characterization of instrumental motivation as efficiency- and goal-oriented. Interestingly, while Hsu [58] reported a negative relationship between extrinsic motivation and PE ($\beta = -0.34$, $p < 0.001$), suggesting that highly extrinsically motivated individuals may neglect usability in the absence of tangible rewards, the current study found the opposite: instrumental motivation positively and significantly predicted PE ($\beta = 0.364$, $p < 0.001$), which means learners with instrumental goals actively seek to use technology as a means to achieve concrete benefits.

Specifically, the key difference between Hsu's negative findings and the positive results of the present study lies in the "certainty of outcomes." In Hsu's study, the external rewards associated with motivation (e.g., "enhancing interest") were ambiguous and non-mandatory, leading highly motivated individuals to potentially disregard perceived ease of use (PE). In contrast, in the present study, the instrumental motivation (INS) is linked to clear and mandatory outcomes (e.g., passing the College English Test), requiring students to "achieve goals efficiently." Consequently, they actively choose tools that are easy to use (high PE) to minimize effort, consistent with Vroom [89], which posits that when outcomes are certain and important, individuals pay greater attention to the efficiency of tools (PE).

It is also noteworthy that the total effect estimates showed a markedly stronger overall influence of INS on IU (TE = 0.160, 0.249, 0.144; $p < 0.001$) compared to INT (TE = 0.031; $p < 0.005$). When considered alongside the finding that INS scores were higher than INT scores (see Table 2), it can be inferred that the intention to use Gen-AI chatbots for English learning is primarily driven by instrumental motivation. In the present study, while INS primarily drives higher Gen-AI usage, it also carries the risk of "instrumental dependence": students may rely on Gen-AI to directly generate answers to fulfill short-term tasks rather than actively engage with language rules, resulting in increased tool dependency accompanied by weakened autonomous learning abilities. From a positive-psychology standpoint, however, such short-term utilitarian or extrinsic

stimulus–driven motivation is not ideal. Although elevated levels of this motivation may boost learner engagement, they can also create an imbalance in intrinsic experience and may even render the learning process aversive [14]. In fact, students with low intrinsic motivation often exhibit a lack of enthusiasm for learning, passive engagement, and reliance on others, and they are generally unwilling to take responsibility for assigned tasks. When faced with learning materials, they tend to show resistance or even express complaints [10]. Consequently, these students lack the drive for autonomous learning and are unlikely to actively use Gen-AI as a genuine tool for English learning, instead engaging with it in a more passive or instrumental manner.

Based on the above discussion, by integrating instrumental and integrative motivation with perceived ease of use, perceived usefulness, and behavioral intention, this study validates the interactive dynamics of the two models in the context of Gen-AI–assisted language learning, thereby providing a replicable framework for future academic research. Second, within an IDLE context and focusing on non-English-major undergraduates at Chinese private universities, distinct patterns emerged in the motivational and perceptual pathways compared to the previous research. This suggests that researchers should re-examine the interplay between TAM and motivational structures across differing conceptual motivational variables, technologies, and sample backgrounds. Third, although integrative motivation levels in this group were relatively low, constructing a positive feedback loop through perceived usefulness can still enhance their healthy and positive willingness to use the Gen-AI tools.

Therefore, for educators, instructional design should address both the efficiency needs of INS and the long-term value of INT. Specifically, Gen-AI chatbots can be embedded in authentic learning tasks—such as batch grammar exercises, essay structure optimization, and simulated oral dialogues—using a "PE–PU dual enhancement" approach [90]. This not only satisfies students' efficiency-driven goals for exams and career preparation but also integrates cultural elements, guiding learners to naturally perceive the interplay of language and culture within tasks [91]. Additionally, establishing a visualized feedback mechanism allows students to compare their work before and after AI assistance, enhancing their trust in the long-term benefits of Gen-AI and sustaining their intention to use it [92].

For system designers, functionality should support both motivational pathways. Cultural recommendation features, cross-cultural dialogue simulations, and cultural competence growth modules can fill the INT→PE gap, enabling learners to perceive both usefulness and ease of use in cultural learning. For INS-driven users, "one-click modes" and learning time estimation features can strengthen the efficiency experience [93]. Coupled with learning-tracking and growth-report functions, these measures make progress visible, transforming "short-term usefulness" into "long-term value."

Finally, at the institutional and policy levels, differentiated strategies should be formulated based on university type and student motivational profiles [94]. For students exhibiting high instrumental but low integrative motivation, curricula that combine utilitarian objectives with cultural awareness could be developed [95], leveraging Gen-AI to generate target-language cultural content and thereby subtly enhance intrinsic drive while reducing reliance on extrinsic, exam-driven motivation [96].

6 Conclusion

6.1 Summary

This study, grounded in TAM and SEMSLM, examined how instrumental and integrative motivations shape non-English-major undergraduates' intentions and perceptions when using Gen-AI chatbots for English learning at Chinese private universities. The findings not only confirm TAM's applicability in AI-mediated language learning within IDLE but also reveal distinct adoption pathways driven by different motivational orientations. The core findings are as follows:

First, participants exhibited a pronounced preference for domestically developed Gen-AI chatbots—an observation that diverges from prior research, which has predominantly emphasized foreign platforms, such as ChatGPT. This underscores the importance of selecting Gen-AI systems with high local usage rates or strong user preference to ensure external validity and contextual relevance in future studies. Second, instrumental motivation proved more salient—and exhibited less variance—than integrative motivation. This finding supports earlier concerns that many private-university undergraduates, constrained by limited language abilities and learning experiences, lack interest in the target language or its cultural contexts, instead regarding English primarily as a means to achieve short-term, utilitarian objectives.

Finally, and perhaps most importantly, SEM indicated that although the paths among the three core variables of the classic TAM model were confirmed, neither type of motivation directly predicted IU. Additionally, it was found that INT had no significant effect on PE and influenced IU only indirectly through PU, suggesting that learners driven by integrative motives prioritize the tool's substantive effectiveness over its ease of use. By contrast, instrumental motivation influenced intention indirectly through both PE and PU—with a stronger influence on PE—suggesting that learners driven by instrumental considerations tend to perceive Gen-AI chatbots as more user-friendly and are more inclined to rely on such external technology to compensate for English-learning challenges.

Although PU's direct effect on IU ($\beta = 0.507$) surpassed that of PE ($\beta = 0.332$)—indicating that students generally require assurance of a tool's practical value in improving English proficiency before developing a strong intention to use it—the overall path analysis revealed that the total effect of instrumental motivation on behavioral intention markedly exceeded that of integrative motivation. It indicates that, for this student cohort, the primary driver of Gen-AI chatbot adoption in English learning remains a pragmatic, results-oriented motivation rather than a deep interest in the language or its culture.

6.2 Limitations

Several limitations of the present study should be acknowledged. First, the sample was limited to non-English-major undergraduates at private universities in China, constraining the generalizability of the findings. Second, by treating Gen-AI chatbots as a unified category and English learning as an undifferentiated construct, the study did not distinguish among specific chatbot brands or discrete language skills (e.g., listening, speaking, reading, writing). Consequently, it remains unclear how particular tool functions or task types may influence learners' acceptance and behavioral pathways; on the other hand, because the questionnaire was finalized at the end of 2024 and data were collected in early 2025, it did not include the domestically popular Chinese Gen-AI tool DeepSeek,

which rose to prominence in February 2025. This highlights the tension between the rapid iteration of Gen-AI technologies and the static nature of measurement instruments, potentially limiting a comprehensive representation of brand preferences and usage intentions. Third, the research was conducted within an IDLE context; findings may not fully translate to formal classroom settings, where instructional goals and motivational dynamics differ significantly. Fourth, a single-administered cross-sectional survey was employed, and this approach did not capture dynamic changes in learners' motivation and behavioral intention over time, thus limiting causal inference due to the absence of longitudinal evidence.

6.3 Future research directions

To address the above limitations, future research could proceed in several directions. First, it should incorporate participants from national key universities, vocational institutions, and English-major cohorts to compare motivational and TAM-path differences across diverse learner backgrounds, thereby enhancing generalizability. Second, variables could be further refined: the characteristics of domestic versus international Gen-AI brands could be distinguished, and brand options updated periodically to reflect technological iteration; meanwhile, the differential effects of specific modules—writing, speaking, and translation—on motivation and perceptions could be examined, with attention to fine-grained impacts on listening, speaking, reading, and writing. Third, deeper investigation into Gen-AI integration in formal classroom environments is warranted, combining teacher and learner assessments with authentic evaluations and survey data to explore underlying mechanisms linking usage and learner motivation. Finally, employing longitudinal designs—such as periodic measurements—alongside qualitative methods (e.g., interviews, observations) would enable examination of the internal mechanisms driving changes in motivation and behavioral intention, thereby enhancing the explanatory power and generalizability.

Appendix: Research Instrument

SEMSLM Part

Please respond using the 5-point Likert scale, where 1 = "Strongly Disagree" and 5 = "Strongly Agree". (请依据5点李克特量表完成作答, 1代表"非常不同意", 5代表"非常同意").

Integrative Motivation Scale [65].

Item	Statement (with Chinese-translated Version)
INT1	I decided to study English because I am interested in English popular culture 我对国外流行文化感兴趣, 所以我想学英语。
INT2	Studying English is important because it will enable me to better understand the way of life in the UK 学英语挺重要的, 比如让我知道不同国家有不同的生活方式。
INT3	Studying English is important for me because it will enable me to appreciate Anglo—American art and literature 学习英语对我来说很重要, 因为它能让我欣赏英语的文艺作品。
INT4	I decided to learn this language in order to better understand the English people way of life 学好英语, 可以更好了解英语母语者的文化。
INT5	I decided to learn English so that I can get to know its speakers better 学好英语可以让我更好地了解老外说的话。

Instrumental Motivation Scale [65].

Item	Statement (with Chinese-translated Version)
INS1	Knowledge of the language can be useful for my further studies, such as at the master's or PhD level 如果我打算读硕士或博士, 掌握英语很有帮助。
INS2	Knowledge of the language will increase my job opportunities 掌握英语让我更好找到工作。
INS3	Proficiency in English can bring me some part—time job opportunities (such as translation or tutoring, etc.) 英语学得好可以给我带来一些兼职机会 (如兼职翻译或家教) 。
INS4	Knowledge of the language will help me when I travel abroad 掌握英语很有用, 比如出国旅行的时候。
INS5	Knowledge of the language will be useful for my future career 英语在我之后的职场生活会很有用。

TAM Part

Please respond using the 7-point Likert scale, where 1 = "Strongly Disagree" and 7 = "Strongly Agree". (请依据7点李克特量表完成作答, 1代表"非常不同意", 7代表"非常同意").

Perceived Usefulness Scale [41].

Item	Statement (with Chinese-translated Version)
PU1	Using Gen—AI chatbot would enable me to accomplish English learning tasks more quickly 1. 使用"人工智能"可以让我更快地完成英语学习任务。
PU2	Using Gen—AI chatbot would improve my English performance 2. 使用"人工智能"可以提高我的英语表现。
PU3	Using Gen—AI chatbot would increase my English learning productivity 3. 使用"人工智能"学习英语可以提高我的学习效率。
PU4	Using Gen—AI chatbot would enhance my English learning effectiveness 4. 使用"人工智能"学习英语可以增强我的学习效果。
PU5	Using Gen—AI chatbot would make it easier to learn English 5. 使用"人工智能"学习英语可以使我的英语学习更轻松。
PU6	I would find Gen—AI chatbot useful in learning English 6. 我认为"人工智能"对我学习英语有用。

Perceived Ease of Use Scale [41].

Item	Statement (with Chinese-translated Version)
PE1	Learning to operate Gen-AI chatbot would be easy for me 1. 学习如何使用"人工智能"对我来说很容易。
PE2	I would find it easy to get Gen-AI chatbot to do what I want it to do 2. 我觉得让"人工智能"按我的指令工作很容易。
PE3	My interaction with Gen-AI chatbot would be clear and understandable 3. 我与"人工智能"的互动内容清晰易懂。
PE4	I would find Gen-AI chatbot to be flexible to interact with 4. 我觉得"人工智能"的应变能力强, 很灵活。
PE5	It would be easy for me to become skillful at using Gen-AI chatbot 5. 熟练使用"人工智能"对我来说很容易。
PE6	I would find Gen-AI chatbot easy to use 6. 我觉得"人工智能"容易使用。

Intention to Use Scale [40].

Item	Statement (with Chinese-translated Version)
IU1	I am willing to spend time and effort to learn how to use Gen—AI chatbot better 我愿意花时间和精力来更好地学习使用"人工智能"。
IU2	I expect to use Gen—AI chatbot frequently in the future for learning English 我预计以后会常使用"人工智能"来学习英语。
IU3	I plan to use Gen—AI chatbot for my learning needs in the future 我计划将来会使用"人工智能"来学习。

Author contributions

Sun Kun: Research execution, field data collection, data organization and analysis. Chew Fong Peng: Supervision, guidance provision, and review. Lau Su Kia: Supervision and guidance. Jing Qiaoqiao: Text polishing and editing.

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Data availability
Data will be made available on request.

Declarations

Ethics approval and consent to participate
The research project has been approved by the Universiti Malaya Research Ethics Committee (UMREC) with Reference Number: UM.TNC2/UMREC_4176. All procedures involving human participants were conducted in accordance with the ethical standards of the institutional committee and with the principles outlined in the Declaration of Helsinki.

Consent for publication
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The authors declare no competing interests.

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